XRPL Validator — Daily Engineering Report · 2026-09-02

Period: 2026-09-02 00:00 to 24:00 US Eastern (2026-09-02 04:00 to 2026-09-03 04:00 UTC); ledgers #106,703,515 to #106,725,810. Every count from the ledger and the validator's logs stops at #106,725,810. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 144 engine receipts recorded (29 fixed, 0 explained, 115 without a recorded fix).
- **Changes:** 0 deployments, 35 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-09-02): 0.96888 (29 missed of 932). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 1 transaction-level difference recorded this day, across 1 ledger. These were not triaged into receipts.

Ledgers with a receipt in the period (#106,703,515–#106,725,810), by step:

Step	Ledgers with a receipt
Step 1 · native engine	144
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#106,703,533	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 46c90df, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-02	—	#106,703,535	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 46c90df, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-03	—	#106,703,546	Step 1 · native engine	2 change(s) of ours the network did not make; 7 object(s) differed in content	no disposition recorded
R-04	—	#106,703,547	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-05	—	#106,703,565	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 object(s) differed in content	fixed: commit d180b37, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-06	—	#106,703,567	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-07	—	#106,703,568	Step 1 · native engine	6 object(s) differed in content	no disposition recorded
R-08	—	#106,703,571	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-09	—	#106,703,584	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-10	—	#106,703,588	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 3 object(s) differed in content	no disposition recorded
R-11	—	#106,703,606	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-12	—	#106,703,622	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-13	—	#106,703,638	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-14	—	#106,703,644	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 3 object(s) differed in content	no disposition recorded
R-15	—	#106,703,646	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-16	—	#106,703,652	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-17	—	#106,703,662	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 object(s) differed in content	no disposition recorded
R-18	—	#106,703,684	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-19	—	#106,703,700	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-20	—	#106,703,712	Step 1 · native engine	2 transaction result(s) differed from rippled's; 11 change(s) of ours the network did not make; 2 object(s) differed in content	no disposition recorded
R-21	—	#106,703,758	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-22	—	#106,703,766	Step 1 · native engine	1 transaction result(s) differed from rippled's; 4 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-23	—	#106,703,769	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-24	—	#106,703,770	Step 1 · native engine	1 transaction result(s) differed from rippled's; 2 change(s) of ours the network did not make; 3 object(s) differed in content	no disposition recorded
R-25	—	#106,703,773	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-26	—	#106,703,840	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 object(s) differed in content	no disposition recorded
R-27	—	#106,703,874	Step 1 · native engine	13 object(s) differed in content	no disposition recorded
R-28	—	#106,704,301	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-29	—	#106,704,302	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-30	—	#106,704,335	Step 1 · native engine	2 object(s) differed in content	no disposition recorded

ID	Time	Ledger(s)	Step	Description	Disposition
R-31	—	#106,704,369	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-32	—	#106,704,453	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-33	—	#106,704,497	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-34	—	#106,704,625	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-35	—	#106,704,718	Step 1 · native engine	2 object(s) differed in content	fixed: commit 1e7c9e9, deployed in cycle 121 (2026-09-11 17:39 UTC)
R-36	—	#106,704,731	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-37	—	#106,704,754	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-38	—	#106,704,755	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-39	—	#106,704,760	Step 1 · native engine	2 object(s) differed in content	no disposition recorded
R-40	—	#106,704,888	Step 1 · native engine	2 object(s) differed in content	fixed: commit 5557187, deployed in cycle 121 (2026-09-11 17:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-02-01, INC-01 is INC-2026-09-02-01. Times are not shown where the reference node no longer holds the ledger.

104 more receipts (#106,705,433–#106,725,324) are in the record and not listed individually; their dispositions: 24 fixed, 80 no disposition recorded.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
04:27 UTC	xrpl-validator	`4bb3198`	fix(ledger): AccountDelete re-arms the destination's password change fee (finding 80)
04:28 UTC	xrpl-validator	`d180b37`	fix(ledger): asfAllowTrustLineClawback needs an empty owner directory (finding 86)
04:30 UTC	xrpl-validator	`6ae1dd1`	fix(ledger): SignerListSet files its entries in account order (finding 83)
04:34 UTC	xrpl-validator	`2ad147b`	fix(ledger): the LP burn rounds the pool balance to nearest (finding 84)
04:37 UTC	xrpl-validator	`bf7ea3e`	fix(ledger): AMMDeposit tfLimitLPToken prices the deposit against EPrice (finding 82)
04:40 UTC	xrpl-validator	`46c90df`	fix(ledger): escrow and channel creation reserve for the new object (finding 85)
04:41 UTC	xrpl-validator	`abb13c0`	fix(ledger): strand remainders are IOUAmounts and the flow ends on the gross budget (finding 81)
04:41 UTC	xrpl-validator	`e64db88`	feat(harness): bundles carry the pool's owner objects for AMM transactions
06:21 UTC	xrpl-validator	`9f513a9`	perf(node): state_replay keeps raw leaves and decodes on read
06:21 UTC	xrpl-validator	`1e7c9e9`	fix(ledger): AMMDeposit tfOneAssetLPToken derives the asset from the tokens (finding 87)
06:39 UTC	xrpl-validator	`5557187`	fix(ledger): AMM maxOffer's 99% cap multiplies nearest, converts down (finding 88)
08:56 UTC	xrpl-validator	`e278926`	fix(ledger): findings 81b, 89, 90, 91 — fill remainders, fee-leg spend, QF limit, saved-in fold
09:17 UTC	xrpl-validator	`7395402`	fix(ledger): CheckCreate refuses a frozen SendMax (finding 92)
10:03 UTC	xrpl-validator	`2610407`	fix(ledger): an in-limited bridged slice floors the XRP mid-leg for a book leg too (finding 93)
11:25 UTC	xrpl-validator	`cece712`	fix(node): ServerVersion and amendment votes ride the voting ledger only
13:35 UTC	xrpl-validator	`a76193e`	fix(scripts): fetch_tx_bundle replays directory pages created earlier in the ledger
13:35 UTC	xrpl-validator	`d4a4421`	fix(ledger): direct IOU payments size the fee step with rippled's mulRatio (finding 96)
13:35 UTC	xrpl-validator	`87383f0`	fix(ledger): tick-size TakerPays multiply rounds half-even to drops (finding 95)
13:35 UTC	xrpl-validator	`20afcdb`	fix(ledger): NFT royalty is waived when the buyer or the seller is the issuer (finding 94)
14:28 UTC	xrpl-validator	`2882024`	fix(ledger): bridge-attempt direct head floors a maker-funds-bound fill (finding 97)
16:47 UTC	xrpl-validator	`e713882`	fix(ledger): NFTokenAcceptOffer refuses own and looped offers (finding 100)
16:47 UTC	xrpl-validator	`5e036e6`	fix(ledger): the pool's spot beating a fee-band tip admits the AMM tail (finding 99)
16:47 UTC	xrpl-validator	`2727985`	test(node): PROBE_LIST dumps every write of a bundle apply
16:47 UTC	xrpl-validator	`989927a`	fix(ledger): a payment's intermediate legs never land on the sender's line (finding 98)
17:48 UTC	xrpl-validator	`fc83c7e`	fix(ledger): AMM tail admission judges the anchored pool offer, not the spot (finding 99, revised)
00:02 UTC	xrpl-validator	`732e144`	fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)
00:02 UTC	xrpl-validator	`e1857fd`	fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)
00:02 UTC	xrpl-validator	`7552fd6`	fix(ledger): a level's pool turn runs before its offers are swept (finding 101)
00:07 UTC	xrpl-validator	`b2f1bc7`	fix(ledger): a level's pool turn runs before its offers are swept (finding 101)
00:07 UTC	xrpl-validator	`6527875`	fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)
00:07 UTC	xrpl-validator	`0bd064e`	fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)
00:55 UTC	xrpl-validator	`878aaa9`	fix(ledger): a book behind an issuer-NoRipple sender line is tecPATH_DRY (finding 105)
02:53 UTC	xrpl-validator	`17eaaac`	fix(ledger): a flow's pools go silent after 30 AMM iterations (finding 107)
03:43 UTC	xrpl-validator	`4c0a834`	fix(ledger): a strand's pool bound prices the slice the pass will consume (finding 109)
03:43 UTC	xrpl-validator	`b884eeb`	fix(ledger): a deposit whose adjusted asset-in reaches zero mints no tokens (finding 108)

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): AccountDelete re-arms the destination's password change fee (finding 80)** (`4bb3198`, xrpl-validator) DeleteAccount::doApply clears the destination's lsfPasswordSpent when it credits a positive balance (DeleteAccount.cpp:436-437), exactly as an XRP Payment does; we kept the bit. #106702154 67271C65: the destination held 0x20010000 and mainnet leaves 0x20000000. Pinned as accountdelete_password_spent_106702154 (red at HEAD~, green here).
- **fix(ledger): asfAllowTrustLineClawback needs an empty owner directory (finding 86)** (`d180b37`, xrpl-validator) SetAccount refuses SetFlag 16 with tecNO_PERMISSION while lsfNoFreeze is set and with tecOWNERS while the owner directory holds anything (SetAccount.cpp:278-292); we set the bit regardless. #106703565 9709366F: an account owning 96 objects — mainnet tecOWNERS, we succeeded. Pinned as accountset_clawback_owners_106703565; the copied vector helpers now honour the bundle's recorded result instead of assuming tesSUCCESS.
- **fix(ledger): SignerListSet files its entries in account order (finding 83)** (`6ae1dd1`, xrpl-validator) SetSignerList sorts the deserialized SignerEntries before writing the list (SetSignerList.cpp:66); we filed them in transaction order. #106703005 F62BBA1C. Pinned as signerlist_sorted_entries_106703005.
- **fix(ledger): the LP burn rounds the pool balance to nearest (finding 84)** (`2ad147b`, xrpl-validator) bump_lp_balance rounded to nearest only when minting and truncated when burning; the subtraction is the same Number arithmetic. #106702692 6F7C52C0 (tfWithdrawAll): 29320.99190061032 - 993.427226328693 rests at 28327.56467428163 on mainnet, ours ended ...62. Pinned as amm_withdraw_all_lp_rounding_106702692.
- **fix(ledger): AMMDeposit tfLimitLPToken prices the deposit against EPrice (finding 82)** (`bf7ea3e`, xrpl-validator) singleDepositEPrice (AMMDeposit.cpp): the full Amount is deposited at Equation 3's tokens when its effective price is within EPrice, otherwise the amount that lands exactly on EPrice is solved from the quadratic (asset rounded up, tokens rounded down, adjustAssetInByTokens after). The mode used to fall through to a 1e7-token placeholder. #106702459 F648B73A: 11.172545 XRP at EPrice 4 drops mints 3320034.6897731 …
- **fix(ledger): escrow and channel creation reserve for the new object (finding 85)** (`46c90df`, xrpl-validator) EscrowCreate judges the post-fee balance against accountReserve(OwnerCount + 1) before anything else — tecINSUFFICIENT_RESERVE for XRP and token escrows alike, then tecUNFUNDED when an XRP amount does not fit above the reserve (Escrow.cpp:496-506); PayChanCreate's preclaim applies the same two tests (PayChan.cpp:205-218). We had no reserve test on either path. #106703533 F84D3EB3 and #106703535 F44C919F: 1000 drops …
- **fix(ledger): strand remainders are IOUAmounts and the flow ends on the gross budget (finding 81)** (`abb13c0`, xrpl-validator) rippled's remainders live in IOUAmount: Number-normalized to 16 digits and zero below an exponent of -96, and the payment loop runs while ONE gross remainder (remainingIn -= f.in) stays positive. Our walk kept every digit and the payment driver broke on a net twin that can carry a one-ULP residue after the strand consumed the whole SendMax. #106703062 AC58204A: 1.335056517376953e-6 BTC spent in full, net residue …
- **feat(harness): bundles carry the pool's owner objects for AMM transactions** (`e64db88`, xrpl-validator) fixAMMv1_1's isOnlyLiquidityProvider walks the AMM account's owner directory counting LP lines; with the other LPs' lines absent a withdraw probe declared the withdrawer the sole LP and refused with tecAMM_INVALID_TOKENS (#106702692 6F7C52C0, live tesSUCCESS). AMM transactions now pull the pool's account_objects and owner directory.
- **perf(node): state_replay keeps raw leaves and decodes on read** (`9f513a9`, xrpl-validator) A full mainnet state replay held ~32 GB of engine JSON (19.8M objects) beside xrpld's 13 GB on the same host, thrashing swap and knocking the upstream into `tracking`. Leaves now hold the serialized blob and `LedgerState::read_json`

decodes through an optional `leaf_decoder` hook (installed by state_replay: decode → hexify addresses → index). JSON leaves (leading `{`) still read as stored, so the engine and every …

- **fix(ledger): AMMDeposit tfOneAssetLPToken derives the asset from the tokens (finding 87)** (`1e7c9e9`, xrpl-validator) rippled's singleDepositTokens (AMMDeposit.cpp) sizes a tfOneAssetLPToken deposit from the requested LPTokenOut: adjust the tokens to the pool's precision, compute the asset in via ammAssetIn for exactly those tokens, and fail with tecAMM_FAILED when that exceeds the offered Amount. Ours sized from the Amount and applied the token cap afterwards, minting a different LP balance and pool root than mainnet.
- **fix(ledger): AMM maxOffer's 99% cap multiplies nearest, converts down (finding 88)** (`5557187`, xrpl-validator) AMMLiquidity::maxOffer sizes the single-path, no-CLOB pool pass as `toAmount(out * Number{99,-2}, Downward)`: the multiply runs in Number's default nearest mode and only the conversion rounds down. `consume` rounded the product down as well, one ulp short whenever the cap binds — a payment draining more than 99% of a pool — and swapAssetOut of the smaller out then moved the in side by ~3e-13. `max_offer_amounts` …
- **fix(ledger): findings 81b, 89, 90, 91 — fill remainders, fee-leg spend, QF limit, saved-in fold** (`e278926`, xrpl-validator) Four engine findings from the 09-02 drill, gated together by HIST60 and WIN98 (the vectors pin each one separately).
- **fix(ledger): CheckCreate refuses a frozen SendMax (finding 92)** (`7395402`, xrpl-validator) CheckCreate::preclaim (rippled CheckCreate.cpp, ungated) rejects a non-XRP SendMax with tecFROZEN when the currency is globally frozen, when the issuer froze the source's trust line (the issuer's side bit), or when the destination froze its own line (the destination's side bit); a party without a line passes. We had no freeze block and created the Check and both directory entries where mainnet claims only the fee.
- **fix(ledger): an in-limited bridged slice floors the XRP mid-leg for a book leg too (finding 93)** (`2610407`, xrpl-validator) The bridged walk rounded the XRP mid-leg up unless leg A was a pool that exhausted the taker's in (finding 51) or the leg-B maker's funds bound the slice (finding 62). rippled's forward pass is in-limited whenever the slice takes the taker's whole remaining in — `limitStepIn` → `ceilInStrict(…, roundUp = false)` → `divRoundStrict(limit, rate, XRP, false)` — for any offer type, so a BOOK leg A floors as well.
- **fix(node): ServerVersion and amendment votes ride the voting ledger only** (`cece712`, xrpl-validator) rippled attaches its server version and its amendment (and fee) votes to the validation of the ledger directly PRECEDING a flag ledger — `Ledger::isVotingLedger()`, seq % 256 == 255 (RCLConsensus.cpp `validate()`). We attached ServerVersion to every validation since 2026-08-29 and amendment votes to the flag ledger itself (seq % 256 == 0), so observers classified both as off-cadence and discarded them.
- **fix(scripts): fetch_tx_bundle replays directory pages created earlier in the ledger** (`a76193e`, xrpl-validator) A page CREATED by an earlier tx of the same ledger was seated from that tx's NewFields (or this tx's FinalFields), which never carry sfIndexes (sMD_Never) — an EMPTY page. q63 509460161CB0@106711585 crossed nothing on it and reported the maker's three objects NOT WRITTEN; with the page's one entry replayed the tx is byte-exact. Route such pages through ensure_page (entries replayed from empty through the earlier …
- **fix(ledger): direct IOU payments size the fee step with rippled's mulRatio (finding 96)** (`d4a4421`, xrpl-validator) `apply_iou_direct` sized the transfer-fee step with `me_muldiv` — a plain ceil for the reverse pass and a plain floor for delivery. rippled's DirectStepI runs IOUAmount `mulRatio` on both passes (DirectStep.cpp:537, :646): the quotient is carried at 18 digits, the IOUAmount constructor normalizes to 16 digits at Number's half-even NEAREST, and only then does a surviving remainder add one ulp for roundUp. For a …

- **fix(ledger): tick-size TakerPays multiply rounds half-even to drops (finding 95)** (`87383f0`, xrpl-validator) `st_multiply`'s XRP side still carried the legacy STAmount form `muldiv(v1, v2, 10^14) + 7`, which is dead under STNumberSwitchover: rippled runs `Number{v1} * Number{v2}` then `XRPAmount{Number}` (STAmount.cpp:1416, canonicalize :880) — the product folded half-even to 16 digits, then half-even to integer drops ("round to nearest, even on tie", Number.h:395). The `+7` breaks every exact half-drop tie upward.
- **fix(ledger): NFT royalty is waived when the buyer or the seller is the issuer (finding 94)** (`20afcdb`, xrpl-validator) rippled pays the NFToken transfer-fee cut only when `seller != issuer && buyer != issuer` (NFTokenAcceptOffer.cpp:423 direct, :542 brokered). All three of our payout helpers tested the seller alone, so an issuer buying its own token back was charged the royalty against the seller and credited with it: #106711435 DC521081D420 (brokered, TransferFee 5 %) moved 500000 drops from the seller to the buyer; #106711449 …
- **fix(ledger): bridge-attempt direct head floors a maker-funds-bound fill (finding 97)** (`2882024`, xrpl-validator) The third fill site in `cross_engine_to_net` — the direct-book head that competes inside a bridge attempt — priced every partial fill with `mul_round16_up`, the `limitOut(…, roundUp = true)` ceil that belongs to a TAKER-bound partial (BookStep.cpp:675). When the MAKER's funds bind, rippled runs the funds clamp instead: `limitOut(…, roundUp = false)` (:790), `mulRoundStrict` under the Number guard at TowardsZero — a …
- **fix(ledger): NFTokenAcceptOffer refuses own and looped offers (finding 100)** (`e713882`, xrpl-validator) rippled's preclaim has three tecCANT_ACCEPT_OWN_NFTOKEN_OFFER rules we never ported: in brokered mode the buy and sell offers may not belong to one account — "the two offers may not form a loop; a broker may not sell the token to the current owner" (NFTokenAcceptOffer.cpp:105-108) — and in either mode an account can't accept an offer it placed itself (:165-167 for a buy offer, :224-226 for a sell offer).
- **fix(ledger): the pool's spot beating a fee-band tip admits the AMM tail (finding 99)** (`5e036e6`, xrpl-validator) An offer crossing's tail turn refused the pool whenever the residual order book tip sat between the net limit and the transfer-rate-inflated one — the #106225714 shape, where the CLOB tip IS the tip, its `qualityUpperBound` carries the input fee and misses the limit. But rippled's `tip()` first anchors the AMM offer on the LOB quality (BookStep.cpp:938-948): when the pool's spot beats the tip that synthetic offer …
- **fix(ledger): a payment's intermediate legs never land on the sender's line (finding 98)** (`989927a`, xrpl-validator) rippled's strand hands a BookStep's output straight to the next step — the issuer's DirectStep or the next book's makers — and the sender's own line in the pass-through currency is never touched. Our hop chain settled every hop against the taker: a hop that is not the first parked its IN on the sender, a hop that is not the last parked its OUT there, and the next hop took it back. Inert on an existing line; on a …
- **fix(ledger): AMM tail admission judges the anchored pool offer, not the spot (finding 99, revised)** (`fc83c7e`, xrpl-validator) Finding 99 admitted the pool whenever its spot beat an order-book tip inside the fee band. Five hundred ledgers later the same account and book showed the rule was too loose: #106715477 01019A7D3E59 and #106715482 52CFB1FDE091 have the spot better than the tip too, yet mainnet rests both offers — "All strands dry" in rippled's trace.
- **fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)** (`732e144`, xrpl-validator) rippled's AMMDeposit::preclaim (featureAMMClawback, AMMDeposit.cpp:255-283) passes BOTH pool assets through requireAuth(WeakAuth) and checkFrozen for the depositor before any funding arithmetic, then the deposited amounts through the STRONG check (:297). requireAuth (RippleStateHelpers.cpp): an issuer with lsfRequireAuth and no line is tecNO_LINE, a line without this side's auth bit tecNO_AUTH; a strong check with …

- **fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)** (`e1857fd`, xrpl-validator) `equal_withdraw_limit` already mirrored rippled's equalWithdrawLimit — lead with Amount and derive Amount2, else lead with Amount2 and derive Amount — but when NEITHER ordering fit both maxima the caller fell through to the verbatim move and paid both amounts in full. Under fixAMMv1_3 rippled returns tecAMM_FAILED there (AMMWithdraw.cpp:949).
- **fix(ledger): a level's pool turn runs before its offers are swept (finding 101)** (`7552fd6`, xrpl-validator) rippled's forEachOffer tries the AMM at the tip's quality BEFORE visiting the level's CLOB offers (BookStep.cpp:837-846): `tryAMM(offers.tip().quality())` executes the anchored pool offer, and when that covers the remaining want execOffer returns false and the tip is never reached — limitSelfCrossQuality never runs on it. Our beyond-limit sweep reaped the level first.
- **fix(ledger): a level's pool turn runs before its offers are swept (finding 101)** (`b2f1bc7`, xrpl-validator) rippled's forEachOffer tries the AMM at the tip's quality BEFORE visiting the level's CLOB offers (BookStep.cpp:837-846): `tryAMM(offers.tip().quality())` executes the anchored pool offer, and when that covers the remaining want execOffer returns false and the tip is never reached — limitSelfCrossQuality never runs on it. Our beyond-limit sweep reaped the level first.
- **fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)** (`6527875`, xrpl-validator) `equal_withdraw_limit` already mirrored rippled's equalWithdrawLimit — lead with Amount and derive Amount2, else lead with Amount2 and derive Amount — but when NEITHER ordering fit both maxima the caller fell through to the verbatim move and paid both amounts in full. Under fixAMMv1_3 rippled returns tecAMM_FAILED there (AMMWithdraw.cpp:949).
- **fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)** (`0bd064e`, xrpl-validator) rippled's AMMDeposit::preclaim (featureAMMClawback, AMMDeposit.cpp:255-283) passes BOTH pool assets through requireAuth(WeakAuth) and checkFrozen for the depositor before any funding arithmetic, then the deposited amounts through the STRONG check (:297). requireAuth (RippleStateHelpers.cpp): an issuer with lsfRequireAuth and no line is tecNO_LINE, a line without this side's auth bit tecNO_AUTH; a strong check with …
- **fix(ledger): a book behind an issuer-NoRipple sender line is tecPATH_DRY (finding 105)** (`878aaa9`, xrpl-validator) rippled's BookStep::check refuses a book that follows a DirectStepI when the line between that step's source and the book's in-issuer is missing (terNO_LINE) or carries NoRipple on the ISSUER's side (terNO_RIPPLE); Payment turns the ter into tecPATH_DRY. Every strand we model opens with the sender -> spend-issuer step and then its first book, so the flag on the issuer's side of the sender's spend line refuses the …
- **fix(ledger): a flow's pools go silent after 30 AMM iterations (finding 107)** (`17eaaac`, xrpl-validator) rippled's AMMContext keeps ONE iteration counter per flow: AMMOffer::consume sets ammUsed_ in either offer mode (fib slice or single-path), flow() calls ammContext.update() once per winning iteration, and AMMLiquidity::getOffer answers nullopt for the rest of the flow once ammIters_ >= kMaxIterations (30). Our AmmFib counted only fib-mode rounds and never capped, and the fib table stopped at 16 entries.
- **fix(ledger): a strand's pool bound prices the slice the pass will consume (finding 109)** (`4c0a834`, xrpl-validator) rippled's AMMLiquidity keeps initialBalances_ from the flow's start and generateFibSeqOffer seeds every fib slice from them; qualityUpperBound and the pass see the same offer. Our consume path records that origin in AmmFib.init at a pool's first use, but the strand upper bound sized its slice from the pool's CURRENT balances, so after a few rounds the bound and the real slice drifted ~1e-4 apart and a near-tie …
- **fix(ledger): a deposit whose adjusted asset-in reaches zero mints no tokens (finding 108)** (`b884eeb`, xrpl-validator) rippled's adjustAssetInByTokens (fixAMMv1_3) recomputes the asset-in for the tokens a deposit earns;

when that rounded-up amount overshoots the deposit, it shrinks the amount by the overshoot, prices the tokens for the shrunken amount and re-adjusts — and a zero (or negative) result is the answer: AMMDeposit refuses tokensAdj == 0 with tecAMM_INVALID_TOKENS, and deposit() refuses a non-positive actual the same way. …

Tests and campaigns

- **test(node): PROBE_LIST dumps every write of a bundle apply** (`2727985`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-02; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

- Receipt R-03 (#106,703,546): open, under investigation.
- Receipt R-04 (#106,703,547): open, under investigation.
- Receipt R-06 (#106,703,567): open, under investigation.
- Receipt R-07 (#106,703,568): open, under investigation.
- Receipt R-08 (#106,703,571): open, under investigation.
- Receipt R-09 (#106,703,584): open, under investigation.
- Receipt R-10 (#106,703,588): open, under investigation.
- Receipt R-11 (#106,703,606): open, under investigation.
- Receipt R-12 (#106,703,622): open, under investigation.
- Receipt R-13 (#106,703,638): open, under investigation.
- Receipt R-14 (#106,703,644): open, under investigation.
- Receipt R-15 (#106,703,646): open, under investigation.
- Receipt R-16 (#106,703,652): open, under investigation.
- Receipt R-17 (#106,703,662): open, under investigation.
- Receipt R-18 (#106,703,684): open, under investigation.
- Receipt R-19 (#106,703,700): open, under investigation.
- Receipt R-20 (#106,703,712): open, under investigation.
- Receipt R-21 (#106,703,758): open, under investigation.

- Receipt R-22 (#106,703,766): open, under investigation.
- Receipt R-23 (#106,703,769): open, under investigation.
- Receipt R-24 (#106,703,770): open, under investigation.
- Receipt R-25 (#106,703,773): open, under investigation.
- Receipt R-26 (#106,703,840): open, under investigation.
- Receipt R-27 (#106,703,874): open, under investigation.
- Receipt R-28 (#106,704,301): open, under investigation.
- Receipt R-29 (#106,704,302): open, under investigation.
- Receipt R-30 (#106,704,335): open, under investigation.
- Receipt R-31 (#106,704,369): open, under investigation.
- Receipt R-32 (#106,704,453): open, under investigation.
- Receipt R-33 (#106,704,497): open, under investigation.
- Receipt R-34 (#106,704,625): open, under investigation.
- Receipt R-36 (#106,704,731): open, under investigation.
- Receipt R-37 (#106,704,754): open, under investigation.
- Receipt R-38 (#106,704,755): open, under investigation.
- Receipt R-39 (#106,704,760): open, under investigation.
- Receipt R-41 (#106,705,433): open, under investigation.
- Receipt R-42 (#106,705,435): open, under investigation.
- Receipt R-44 (#106,706,003): open, under investigation.
- Receipt R-45 (#106,706,976): open, under investigation.
- Receipt R-46 (#106,707,067): open, under investigation.
- Receipt R-47 (#106,707,081): open, under investigation.
- Receipt R-48 (#106,707,206): open, under investigation.
- Receipt R-49 (#106,707,213): open, under investigation.
- Receipt R-50 (#106,707,239): open, under investigation.
- Receipt R-51 (#106,707,273): open, under investigation.
- Receipt R-52 (#106,707,307): open, under investigation.
- Receipt R-53 (#106,707,322): open, under investigation.
- Receipt R-54 (#106,707,341): open, under investigation.
- Receipt R-55 (#106,707,376): open, under investigation.
- Receipt R-56 (#106,707,506): open, under investigation.
- Receipt R-57 (#106,707,542): open, under investigation.
- Receipt R-58 (#106,707,672): open, under investigation.
- Receipt R-59 (#106,708,013): open, under investigation.
- Receipt R-61 (#106,708,738): open, under investigation.
- Receipt R-62 (#106,708,762): open, under investigation.
- Receipt R-64 (#106,709,625): open, under investigation.
- Receipt R-65 (#106,709,803): open, under investigation.

- Receipt R-66 (#106,709,868): open, under investigation.
- Receipt R-67 (#106,709,901): open, under investigation.
- Receipt R-68 (#106,709,902): open, under investigation.
- Receipt R-69 (#106,709,908): open, under investigation.
- Receipt R-70 (#106,709,911): open, under investigation.
- Receipt R-71 (#106,709,922): open, under investigation.
- Receipt R-72 (#106,710,229): open, under investigation.
- Receipt R-73 (#106,710,257): open, under investigation.
- Receipt R-74 (#106,710,261): open, under investigation.
- Receipt R-75 (#106,710,298): open, under investigation.
- Receipt R-76 (#106,710,330): open, under investigation.
- Receipt R-77 (#106,710,383): open, under investigation.
- Receipt R-79 (#106,710,959): open, under investigation.
- Receipt R-80 (#106,711,414): open, under investigation.
- Receipt R-85 (#106,711,981): open, under investigation.
- Receipt R-87 (#106,712,271): open, under investigation.
- Receipt R-88 (#106,712,561): open, under investigation.
- Receipt R-89 (#106,712,630): open, under investigation.
- Receipt R-90 (#106,712,669): open, under investigation.
- Receipt R-92 (#106,712,862): open, under investigation.
- Receipt R-93 (#106,712,863): open, under investigation.
- Receipt R-94 (#106,712,864): open, under investigation.
- Receipt R-95 (#106,712,881): open, under investigation.
- Receipt R-96 (#106,713,277): open, under investigation.
- Receipt R-98 (#106,714,198): open, under investigation.
- Receipt R-99 (#106,714,317): open, under investigation.
- Receipt R-101 (#106,715,134): open, under investigation.
- Receipt R-104 (#106,715,500): open, under investigation.
- Receipt R-105 (#106,715,603): open, under investigation.
- Receipt R-106 (#106,715,622): open, under investigation.
- Receipt R-107 (#106,715,749): open, under investigation.
- Receipt R-108 (#106,716,079): open, under investigation.
- Receipt R-109 (#106,716,436): open, under investigation.
- Receipt R-111 (#106,716,808): open, under investigation.
- Receipt R-112 (#106,716,981): open, under investigation.
- Receipt R-113 (#106,717,380): open, under investigation.
- Receipt R-114 (#106,717,476): open, under investigation.
- Receipt R-115 (#106,717,850): open, under investigation.
- Receipt R-116 (#106,718,379): open, under investigation.

- Receipt R-117 (#106,718,414): open, under investigation.
- Receipt R-118 (#106,718,522): open, under investigation.
- Receipt R-119 (#106,720,543): open, under investigation.
- Receipt R-120 (#106,720,617): open, under investigation.
- Receipt R-122 (#106,720,824): open, under investigation.
- Receipt R-127 (#106,722,535): open, under investigation.
- Receipt R-130 (#106,723,143): open, under investigation.
- Receipt R-132 (#106,723,494): open, under investigation.
- Receipt R-133 (#106,723,672): open, under investigation.
- Receipt R-134 (#106,723,686): open, under investigation.
- Receipt R-135 (#106,724,000): open, under investigation.
- Receipt R-136 (#106,724,002): open, under investigation.
- Receipt R-138 (#106,724,098): open, under investigation.
- Receipt R-139 (#106,724,166): open, under investigation.
- Receipt R-140 (#106,724,186): open, under investigation.
- Receipt R-141 (#106,724,467): open, under investigation.
- Receipt R-142 (#106,724,867): open, under investigation.
- Receipt R-143 (#106,725,248): open, under investigation.
- Receipt R-144 (#106,725,324): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Bounded Concentrated AMM Liquidity** (amm-defi) Add a new AMMLiquidityPosition ledger object (owner-reserved, keylet::ammPosition(ammID, account, seq)) carrying sfTickLower/sfTickUpper price-tick SF
- **Shielded EscrowFinish for Confidential Transfers** (privacy) Add an lsfConfidential flag and sfAmountCommitment (a Pedersen commitment, reusing ConfidentialTransfer's commitment scheme) to Escrow objects created
- **Byte-Weighted Owner Reserves** (fees) Replace the flat per-object OwnerCount increment with sfReserveUnits, a deterministic ceil(serializedSize/128) computed at object creation for RippleS

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-02 04:27	09-02 04:27	xrpl-validator	`4bb3198`	fix(ledger): AccountDelete re-arms the destination's password change fee (finding 80)
09-02 04:28	09-02 04:28	xrpl-validator	`d180b37`	fix(ledger): asfAllowTrustLineClawback needs an empty owner directory (finding 86)
09-02 04:30	09-02 04:30	xrpl-validator	`6ae1dd1`	fix(ledger): SignerListSet files its entries in account order (finding 83)
09-02 04:34	09-02 04:34	xrpl-validator	`2ad147b`	fix(ledger): the LP burn rounds the pool balance to nearest (finding 84)
09-02 04:37	09-02 04:37	xrpl-validator	`bf7ea3e`	fix(ledger): AMMDeposit tfLimitLPToken prices the deposit against EPrice (finding 82)
09-02 04:40	09-02 04:40	xrpl-validator	`46c90df`	fix(ledger): escrow and channel creation reserve for the new object (finding 85)
09-02 04:41	09-02 04:41	xrpl-validator	`abb13c0`	fix(ledger): strand remainders are IOUAmounts and the flow ends on the gross budget (finding 81)
09-02 04:41	09-02 04:41	xrpl-validator	`e64db88`	feat(harness): bundles carry the pool's owner objects for AMM transactions
09-02 06:21	09-02 06:21	xrpl-validator	`9f513a9`	perf(node): state_replay keeps raw leaves and decodes on read
09-02 06:21	09-02 06:21	xrpl-validator	`1e7c9e9`	fix(ledger): AMMDeposit tfOneAssetLPToken derives the asset from the tokens (finding 87)
09-02 06:39	09-02 06:39	xrpl-validator	`5557187`	fix(ledger): AMM maxOffer's 99% cap multiplies nearest, converts down (finding 88)
09-02 08:56	09-02 08:56	xrpl-validator	`e278926`	fix(ledger): findings 81b, 89, 90, 91 — fill remainders, fee-leg spend, QF limit, saved-in fold
09-02 09:17	09-02 09:17	xrpl-validator	`7395402`	fix(ledger): CheckCreate refuses a frozen SendMax (finding 92)
09-02 10:03	09-02 10:03	xrpl-validator	`2610407`	fix(ledger): an in-limited bridged slice floors the XRP mid-leg for a book leg too (finding 93)
09-02 11:25	09-02 11:25	xrpl-validator	`cece712`	fix(node): ServerVersion and amendment votes ride the voting ledger only
09-02 13:35	09-02 13:35	xrpl-validator	`a76193e`	fix(scripts): fetch_tx_bundle replays directory pages created earlier in the ledger
09-02 13:35	09-02 13:35	xrpl-validator	`d4a4421`	fix(ledger): direct IOU payments size the fee step with rippled's mulRatio (finding 96)
09-02 13:35	09-02 13:35	xrpl-validator	`87383f0`	fix(ledger): tick-size TakerPays multiply rounds half-even to drops (finding 95)
09-02 13:35	09-02 13:35	xrpl-validator	`20afcdb`	fix(ledger): NFT royalty is waived when the buyer or the seller is the issuer (finding 94)
09-02 14:28	09-02 14:28	xrpl-validator	`2882024`	fix(ledger): bridge-attempt direct head floors a maker-funds-bound fill (finding 97)
09-02 16:47	09-02 16:47	xrpl-validator	`e713882`	fix(ledger): NFTokenAcceptOffer refuses own and looped offers (finding 100)
09-02 16:47	09-02 16:47	xrpl-validator	`5e036e6`	fix(ledger): the pool's spot beating a fee-band tip admits the AMM tail (finding 99)
09-02 16:47	09-02 16:47	xrpl-validator	`2727985`	test(node): PROBE_LIST dumps every write of a bundle apply
09-02 16:47	09-02 16:47	xrpl-validator	`989927a`	fix(ledger): a payment's intermediate legs never land on the sender's line (finding 98)
09-02 17:48	09-02 17:48	xrpl-validator	`fc83c7e`	fix(ledger): AMM tail admission judges the anchored pool offer, not the spot (finding 99, revised)

Committed	Merged into main	Repo	Commit	Subject
09-03 00:02	09-03 00:02	xrpl-validator	`732e144`	fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)
09-03 00:02	09-03 00:02	xrpl-validator	`e1857fd`	fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)
09-03 00:02	09-03 00:02	xrpl-validator	`7552fd6`	fix(ledger): a level's pool turn runs before its offers are swept (finding 101)
09-03 00:07	09-03 00:07	xrpl-validator	`b2f1bc7`	fix(ledger): a level's pool turn runs before its offers are swept (finding 101)
09-03 00:07	09-03 00:07	xrpl-validator	`6527875`	fix(ledger): a tfTwoAsset withdraw that fits neither ordering is tecAMM_FAILED (finding 102)
09-03 00:07	09-03 00:07	xrpl-validator	`0bd064e`	fix(ledger): AMMDeposit runs requireAuth and checkFrozen on both pool assets (finding 104)
09-03 00:55	09-03 00:55	xrpl-validator	`878aaa9`	fix(ledger): a book behind an issuer-NoRipple sender line is tecPATH_DRY (finding 105)
09-03 02:53	09-03 02:53	xrpl-validator	`17eaaac`	fix(ledger): a flow's pools go silent after 30 AMM iterations (finding 107)
09-03 03:43	09-03 03:43	xrpl-validator	`4c0a834`	fix(ledger): a strand's pool bound prices the slice the pass will consume (finding 109)
09-03 03:43	09-03 03:43	xrpl-validator	`b884eeb`	fix(ledger): a deposit whose adjusted asset-in reaches zero mints no tokens (finding 108)

Times in UTC. A commit appears on the day it was committed to main or merged into it.